Literature Status: Hypercontractive Homogeneous Cotype

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Source Question

The source paper is Daniel Carando, Andreas Defant, and Pablo Sevilla-Peris, *Some polynomial versions of cotype and applications*, arXiv:1503.00850.

In Section 1, after defining hypercontractive homogeneous cotype, the authors ask whether this property is equivalent to ordinary cotype. In their notation, a Banach space X has hypercontractive homogeneous cotype q if there is $C > 0$ such that for every m and every finitely supported family $(x_\alpha)_{|\alpha|=m} \subset X$,

$$\left(\sum_{|\alpha|=m} \|x_\alpha\|^q \right)^{1/q} \leq C^m \left(\int_{\mathbb{T}^\infty} \left\| \sum_{|\alpha|=m} x_\alpha z^\alpha \right\|^2 dz \right)^{1/2}.$$

They conjecture that, for each $q \geq 2$, a Banach space has this property if and only if it has ordinary Rademacher cotype q.

Answering Literature

The later paper is Daniel Carando, Felipe Marceca, and Pablo Sevilla-Peris, *Hausdorff-Young type inequalities for vector-valued Dirichlet series*, arXiv:1904.00041.

Its introduction explicitly states that the paper proves Rademacher cotype is equivalent to the hypercontractive homogeneous cotype defined in arXiv:1503.00850. The precise theorem is Theorem `cotimplicapolcot` in Section 3. It proves that, for $2 \leq q < \infty$, the following are equivalent:

- X has cotype q;
- there is $C > 0$ such that every X-valued polynomial of degree at most m satisfies

$$\left(\sum_{|\alpha|\leq m} \|x_\alpha\|^q \right)^{1/q} \leq C^m \left(\int_{\mathbb{T}^n} \left\| \sum_{|\alpha|\leq m} x_\alpha z^\alpha \right\|^q dz \right)^{1/q}.$$

Remark `expo` then records that the polynomial L_q norm can be replaced by any L_r norm, with constants still growing only exponentially in m. Restricting this theorem to homogeneous polynomials and using $r = 2$ gives the definition of hypercontractive homogeneous cotype in the source paper. The converse direction is immediate already from the case $m = 1$.

Status

The conjecture from arXiv:1503.00850 is therefore already answered affirmatively by arXiv:1904.00041. This packet is a literature-status packet, not a new proof produced by the run.

References

[1] D. Carando, A. Defant, and P. Sevilla-Peris, *Some polynomial versions of cotype and applications*, arXiv:1503.00850, 2015.

[2] D. Carando, F. Marceca, and P. Sevilla-Peris, *Hausdorff-Young type inequalities for vector-valued Dirichlet series*, arXiv:1904.00041, 2019.